

Worksheet 5

Fall 2026

W 5-1 Consider the following average annual returns:

Investment	Average return
Small stocks	23.2%
S&P 500	13.2%
Corporate bonds	7.5%
Treasury bonds	6.2%
Treasury bills	4.8%

- (a) What is the excess return for the portfolio of small stocks?
- (b) What is the excess return for the portfolio of corporate bonds?
- (c) Estimate the market risk premium.
- (d) Apply the CAPM to determine the beta of the corporate bonds portfolio.
- (e) If the volatility of the market is 20%, determine the volatility of the corporate bonds portfolio.

W 5-2 A portfolio consists of the following investments:

Stock	Proportion of PF	Beta
A	60%	1.50
B	40%	1.25

To lower the portfolio's beta to 0.50, the investor sells a fraction α of the portfolio and invests the proceeds in a risk-free bond. Determine α .

W 5-3 Consider the following information on two securities:

	Volatility	Beta
Asset A	22%	2.1
Asset B	38%	0.7

- (a) Which asset earns a higher risk premium?
- (b) If the risk-free rate is 3% and the market portfolio earns an average return of 9%, what are the expected returns for each of the two assets?
- (c) Suppose you invest in a portfolio that consists of 75% Asset A and 25% Asset B. Determine your portfolio expected return and portfolio beta.

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- (d) Do the same when you invest 150% in Asset A and -50% in Asset B.
 - (e) Visualize your results on a graph with beta on the x-axis and the expected return on the y-axis. Verify that all your answers fall on a straight line.